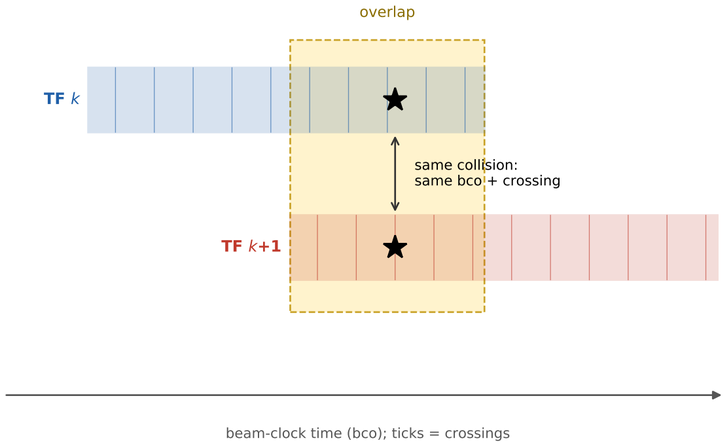
ana581, page 1: the clean sample of events -- whole-event copies, track copies across crossings, tracks from another collision -- schematic, not to scale

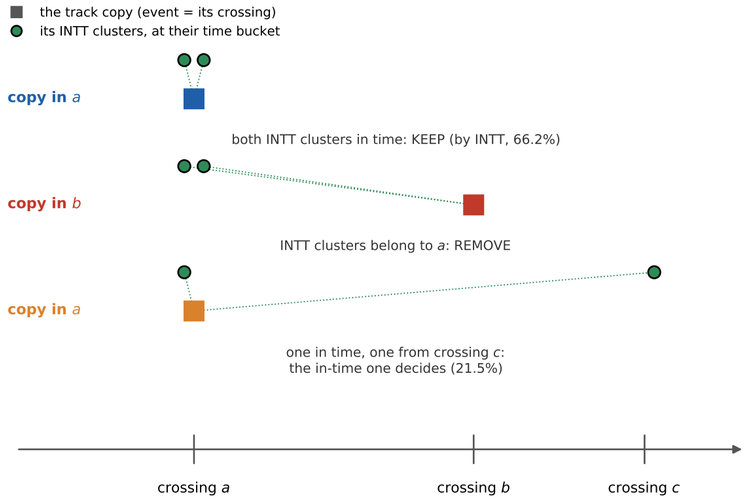
1. One collision in two overlapping trigger frames
TFdup: skip the second copy (whole event)



The copy in TF k+1 is the SAME event. Kept, it would sit in the mixing pool next to its original and mix with its own copy: a like-sign MIXED spike at (dy, dϕ) = (0,0), x7.34 / x7.49 with nothing on (ana578 x7.97).

TFdup skips 13.0% of the ana581 rows (ana578 12.8%), 94% of them one TF later. First copy read is kept. With TFdup: x1.18 / x1.11; with XTF and the mixing exclusion too: 1.02.

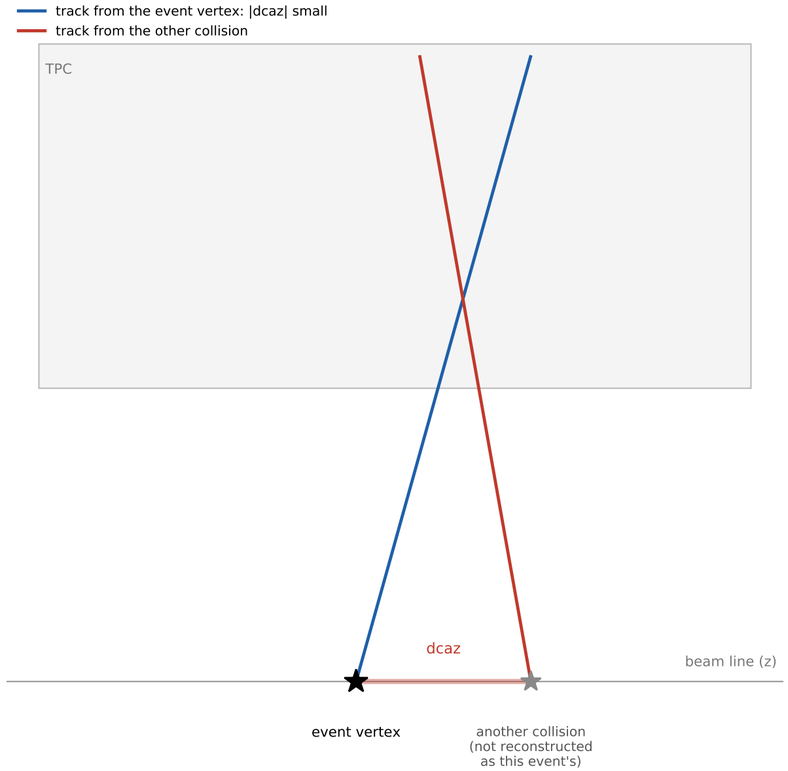
2. One track copied into another crossing of the same TF
XTF: keep the copy whose INTT clusters are in time



The matcher can attach one silicon seed (same Si cluster keys) to TPC tracks in different crossings of one TF: the same particle twice.

ana581: still 26% of the tracks with $\geq$ 2 INTT clusters carry one from an unrelated crossing (ana578 26%). Judged by |dcaz| (10M of 79516): INTT 0.956, in-time 0.946, quality 0.923 right -- 0.3-1.3% better than ana578 in every class; in-time beats quality on the same pairs, so the order stays. 238M duplicate pairs (7 runs): INTT 66.2%, in-time 21.5%, quality 6.6%, neither in time 5.7% (both dropped); 235M tracks removed.

3. A track attached to the wrong vertex
level-0 cut |dcaz| $\leq$ 1.5 cm (side view, r vs z)

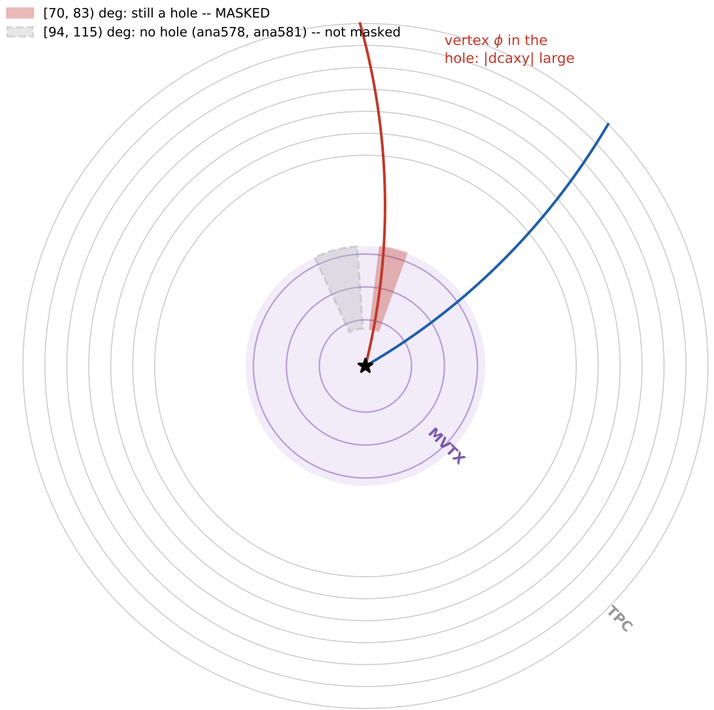


Raw XingView: jet- and photon-triggered TFs have more tracks per event near the trigger (|crossing| ~ 30; jet - MB +0.085 ± 0.017 at crossings -20..-5).
N-1 test per cut (20M rows of 79516): the event cuts do not touch the bump; with every track cut but dcaz it stays (+0.085), with dcaz it is gone (-0.010 ± 0.015).
So the bump is tracks with |dcaz| > 1.5 cm: from other collisions near the trigger, removed by the standard level-0 cut. All crossings above 2 are usable.
This was the ana578 test; not redone on ana581 (needs AllX files; skipped, user 11-02).

Event = one crossing with a vertex; TF = trigger frame (the block of streaming data that holds many crossings). Numbers (7-run production ana581: TFdup, XTF counts; run 79516: README_SplitTracks581 sec 2; panel 3: README_SplitTracks578 sec 1.7).

ana581, page 2: Area A, single-track acceptance -- holes that change from event to event, masked per track -- schematic and data

4. A1 -- the SiPhi zones: silicon holes in vertex ϕ mask [70, 83) deg only (SiPhiMask)



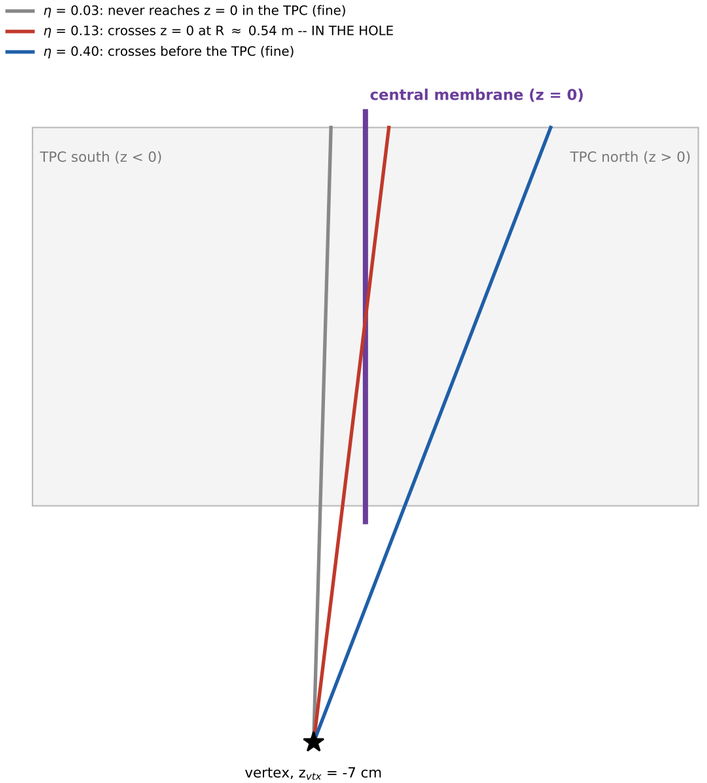
Tracks per 1-deg bin of vertex ϕ and their <|dcaxy|> (level-0 tracks, SiPhi mask off; ana581_ZB, 2.85e9 / 2.67e9 tracks per charge):

[70,83): 2.22% / 2.31% of the tracks (flat 3.61%), <|dcaxy|> 1.70 / 1.63x the median (ana578 2.19 / 2.32%, 1.63 / 1.59x): sharp in vertex ϕ, both charges, a defect at the silicon radius -- as in ana578.

[94,115): 5.69 / 5.92% (flat 5.83%), <|dcaxy|> normal (1.09-1.12x): no hole.

The mask costs 2.26% of the ana581 tracks (ana578 2.3%). Holds.

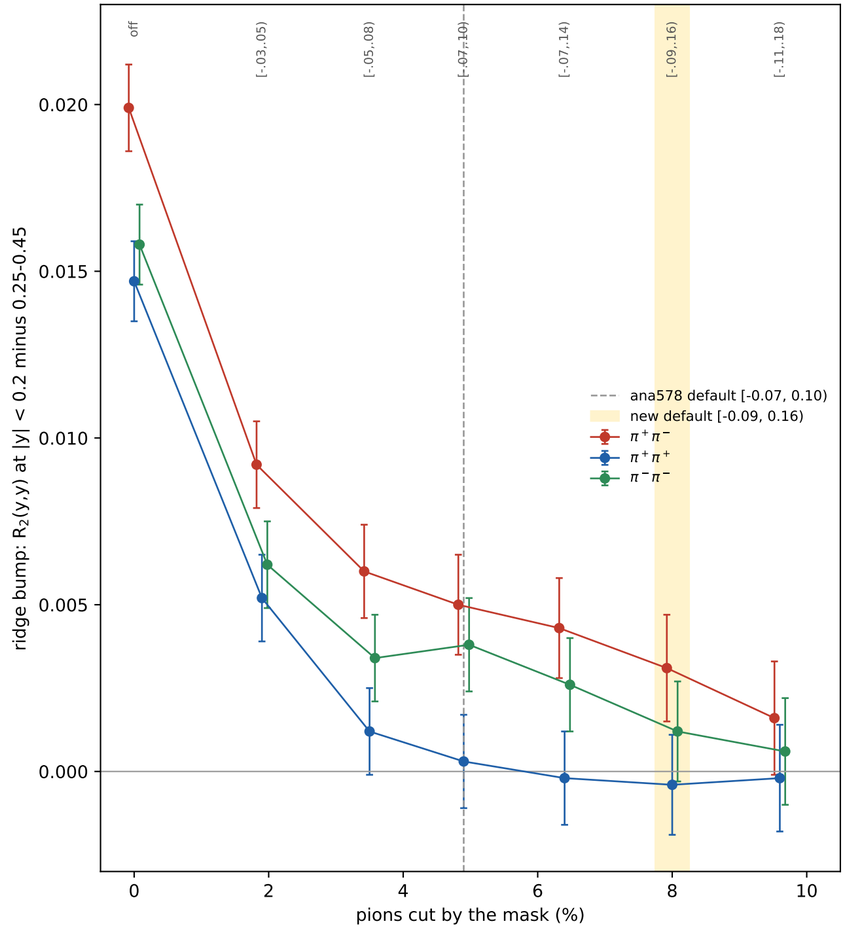
5. A2 -- the TPC central-membrane hole tracks that cross z = 0 inside the TPC (CM mask, wider in ana581)



Side view (z stretched x5): a track crosses z = 0 at R = -z_{vtx}/sinh η; for R in the TPC its clusters sit on both sides of the membrane: a band in η that moves with z_{vtx}. w = (η + 0.018 z_c) sign(-z_c), z_c = the z_{vtx}-slice center: the hole sits at w $\approx$ 0.

ana581: same place, shallower again (minimum 0.52 of the plateau; ana578 0.42). A hole that is the same in every event cancels in S/M; this one opens and shuts from event to event, so it makes a bump in $R_2(y_1,y_2)$ at $y_1 = y_2 \sim 0$.
Mask: w in [-0.09, 0.16), fixed per 2-cm slice: 8.0% of the pions (ana578 [-0.07, 0.10), 4.3%).

6. A2 decided after Area B: the bump vs the mask width all of 79516 (152M events), every B cleaner on

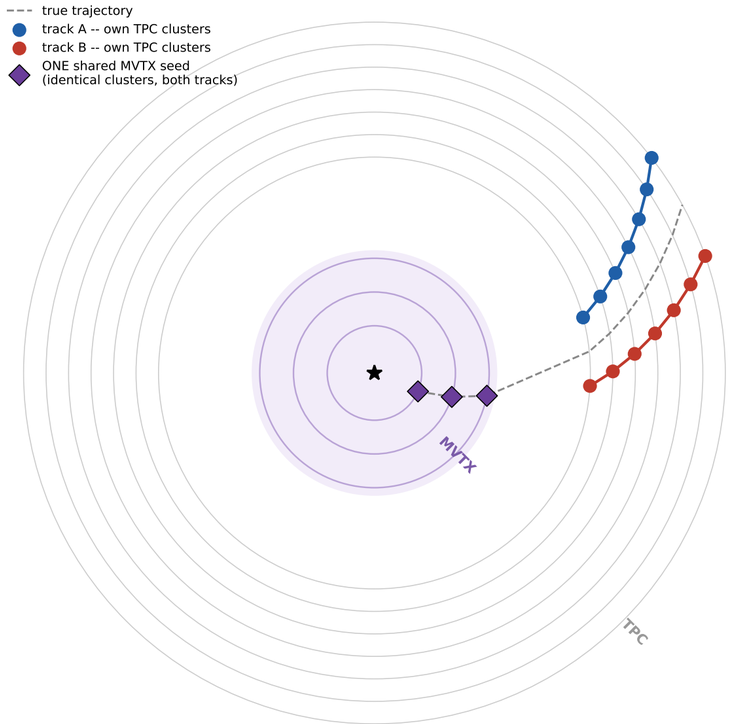


Without the mask a CHARGE-INDEPENDENT bump of 0.015-0.020 (12-16σ) is left after Area B (ana578 0.021-0.027); the mask removes it, monotonically with the width. $\pi^+\pi^+$ is gone from [-0.05, 0.08); $\pi^-\pi^-$ and $\pi^+\pi^-$ keep falling slowly to the widest window.
Decision (user, 11-03): [-0.09, 0.16), 8.0% of the pions: 0.0031 / -0.0004 / 0.0012.
The widths are nested (same events): neighboring points are correlated.

Numbers: README_SplitTracks581 secs A1 (SiPhi_ana581.pdf), A2 (CMHole_ana581.pdf) and the A2 width scan (ST581_A2_CMMask.pdf); stat errors from the same-event bins.

ana581, page 3: Area B, like-sign split tracks -- one silicon seed inside two same-charge tracks -- schematic and data

7. Like-sign "duplicate" in the pregate: SKF ≥ 0.10 AND SL < 0.18 (path 1), or ≥ 2 MVTX (path 3)



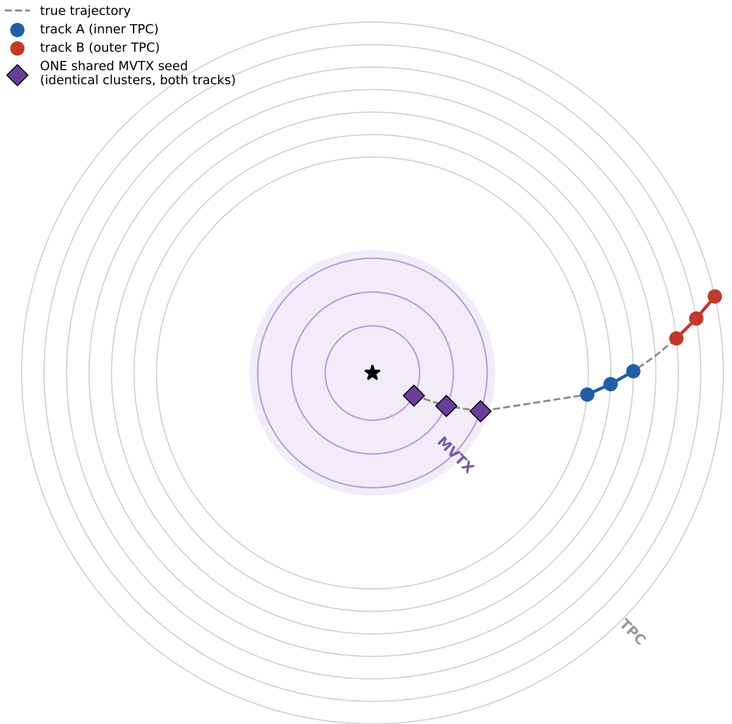
Same charge, same silicon seed, two TPC cluster sets on the SAME layers (different clusters). SL ≈ -0.45, RadialGap ≈ 0.02.

The fragments share the seed but can differ in p_T: they land at $(\Delta\eta,\Delta\phi) \approx (0, \Delta\phi_0)$; the pregate is centered there (panel 9).

ana581 (10M of 79516, B3): SL does 93% of the LS removals. Path 3 (≥ 2 MVTX layers with the same cluster) is NEEDED: without it 30.3k more tracks stay and the like-sign bump comes back (+0.047 $\pi^+\pi^+$, +0.024 $\pi^-\pi^-$).

π^+ splits 1.77 ± 0.02 times as often as π^- (ana578 1.80). LS paths: 1.021M tracks flagged per 10M events (ana578 1.073M).

8. Like-sign "complementary" in the pregate: SKF ≥ 0.10 AND RadialGap ≥ 0.45 (path 2)

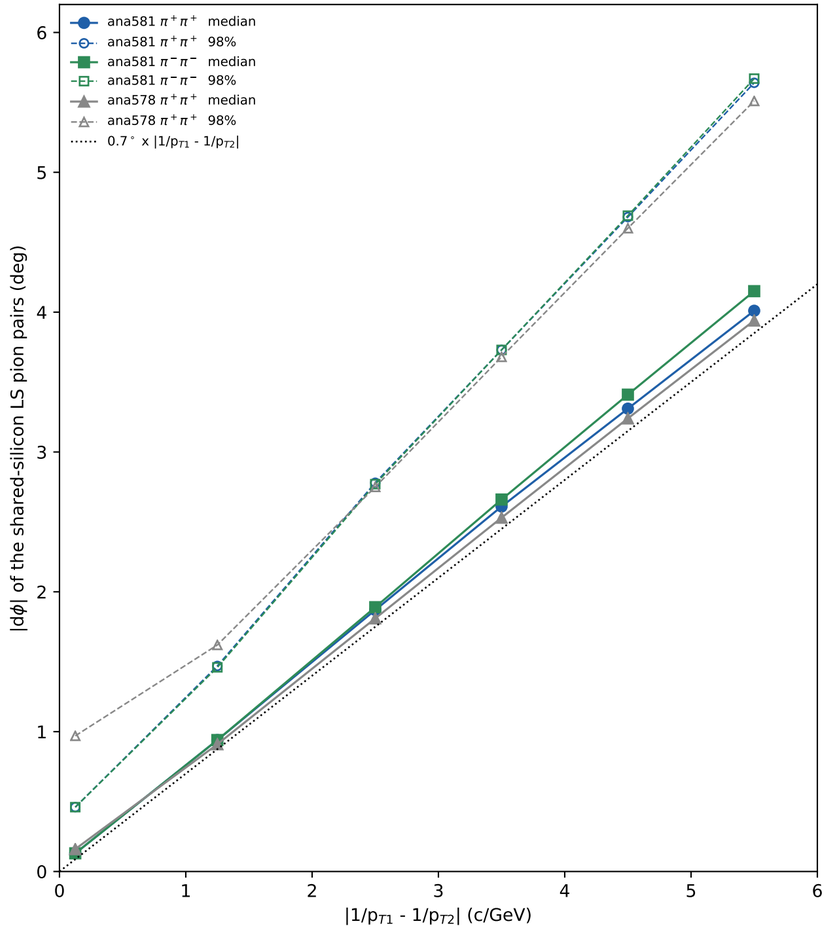


Same charge, same silicon seed, one trajectory's TPC clusters torn into DISJOINT inner/outer segments. SL = +1 (blind), RadialGap ≈ 0.53.

ana581 (10M of 79516): 11,058 RadialGap pairs; with RG off path 3 takes almost all of them -- only 89 more tracks survive and the bump does not move. Only which path gets the pair first (as in ana578: 63).

Nearly redundant; kept (costs nothing).

9. B1 -- the like-sign split band and the pregate shared-silicon LS pion pairs, no cleaner (10M of 79516)

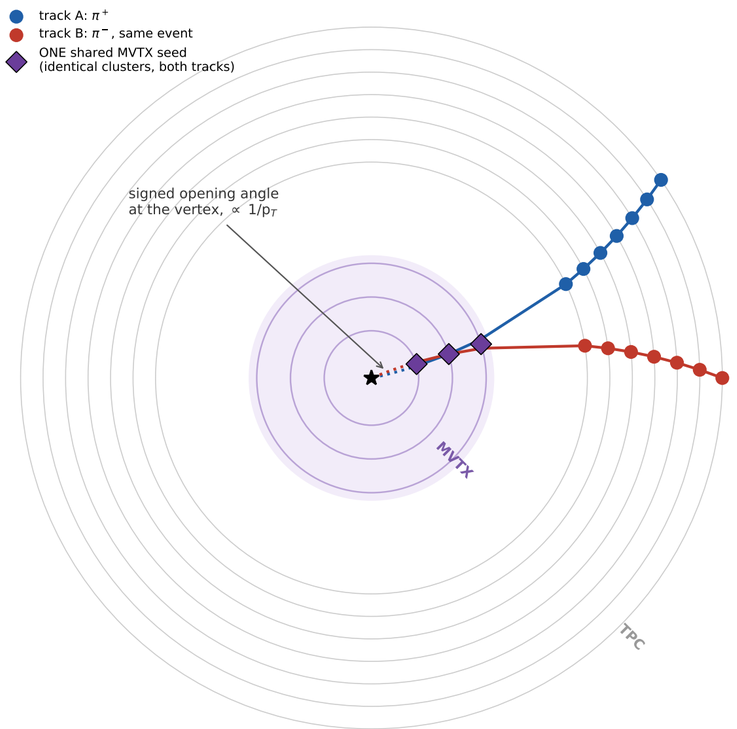


Same law as ana578: $|d\phi| \sim 0.7^\circ \times |1/p_{T1} - 1/p_{T2}|$, both charges. CHANGED: narrower at small curvature difference (98% edge 0.46° below 0.25 c/GeV; ana578 0.97°); above 2 c/GeV as before. Gate-frame core sigma 0.10-0.30° (ana578 0.13-0.34°), well inside w = 2.0-2.6°: 99.65-99.82% of the shared-silicon pairs are inside the pregate (ana578 99.6-99.7%); 8% fewer LS splits per 10M events (518k vs 562k). Decision: the pregate stays (dps20, R = 6 cm, dη 0.040).

Purple diamonds = MVTX clusters; grey circles = TPC (schematic). The proof of "one silicon seed in two tracks" is Si cluster IDENTITY (same physical cluster). Loser rule: fewer TPC clusters, then worse fit quality. Numbers: README_SplitTracks581 secs B1, B3 (10M of 79516).

ana581, page 4: Area B, opposite-sign -- one silicon seed in two opposite-charge tracks, and loopers -- schematic and data

10. B4 -- one silicon seed, two opposite-charge TPC tracks
ULS veto: SiSplitScore ≥ 0.05, any angle



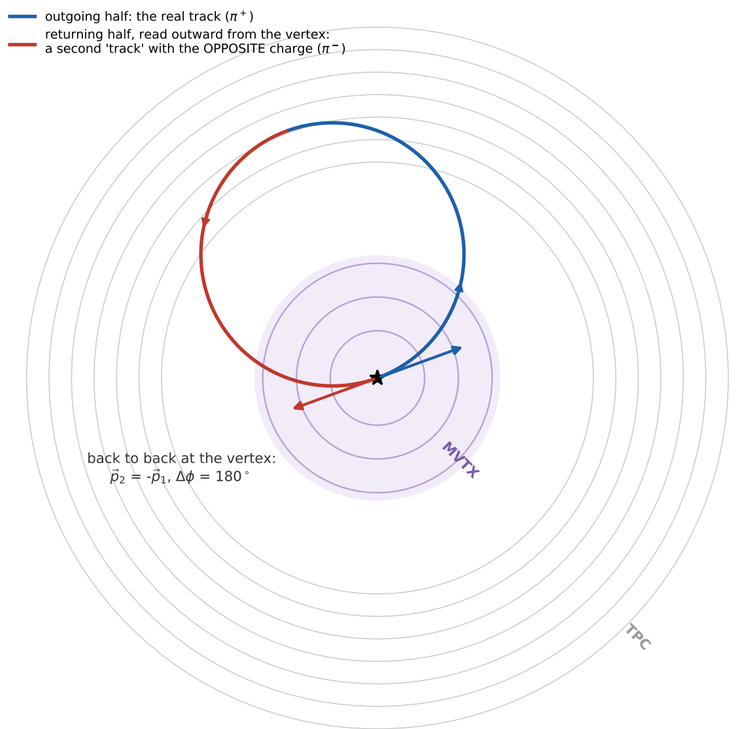
The matcher attaches one silicon seed to two TPC seeds of opposite curvature in the SAME crossing (other crossings of the TF: XTF, page 1).

A $\pi^+\pi^-$ pair at very small Q: the C(Q)/(1+r_2) bump at 20-35 MeV (panel 12) and the unlike-sign ridge bump at y ~ 0 (0.247 -> 0.016 with the veto).

ana581 (10M of 79516): 396,327 pairs flagged (ana578 369,769): 7% more -- the larger $\pi^+\pi^-$ excess at (0,0) of step 2. Like-sign C(Q) unchanged.

Decision: the veto stays as it is.

11. B5 -- a looper: one curling track seen twice
looper veto: remove the worse half



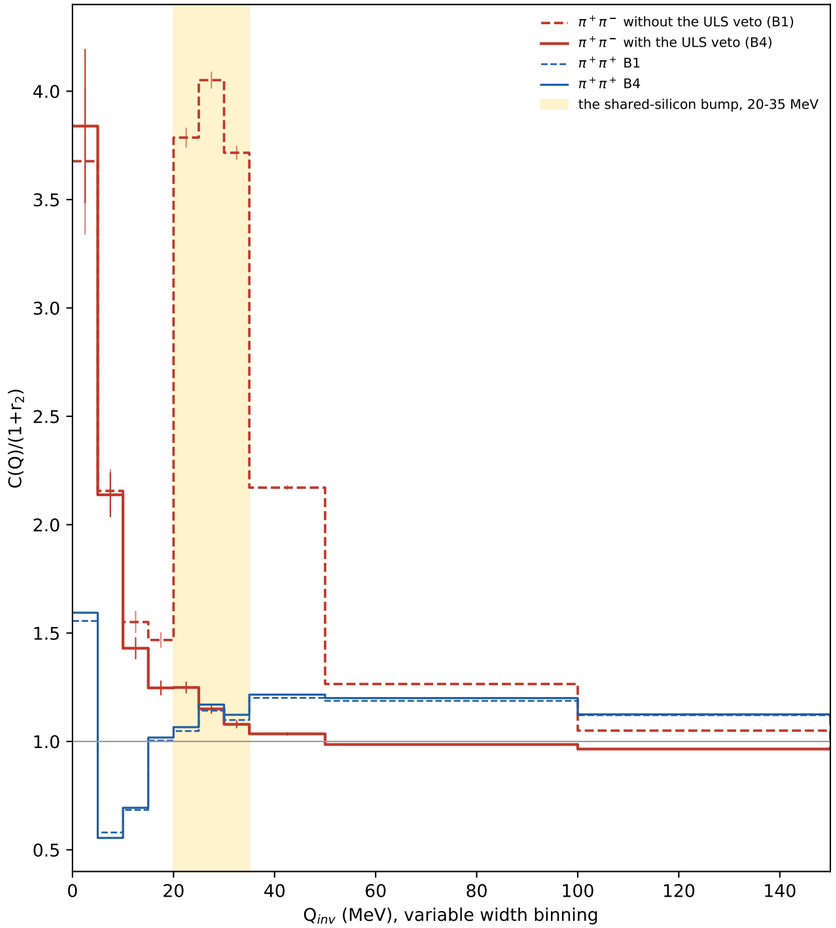
p_T < ~0.17 GeV/c: the circle closes inside the TPC and the track comes back through the center: an opposite-charge 'track', back to back, same |p|.

Veto: OS, both |p| < 0.20 GeV/c, $\Delta\phi \geq 170^\circ$, $|\vec{p}_1+\vec{p}_2|/(|p_1|+|p_2|) < 0.08$.

ana581 (all of 79516): 63.3k excess back-to-back pairs below 0.08 over a control of 28.8k (relative sum 0-0.02: 21.7x; ana578 36x); the veto takes them to 0.89-0.96. ~22% of the excess lies above 0.08 (1.26 at 0.08-0.10): watch.

109k tracks removed (ana578 130k): ~15% fewer loopers.
$\pi^+\pi^-$ away-side spike at dy ~ 0: +0.028 -> +0.009. Decision: 0.08 stays.

12. B4 in the data: $\pi^+\pi^-$ C(Q)/(1+r_2)
10M of 79516 (ana581), without and with the ULS veto

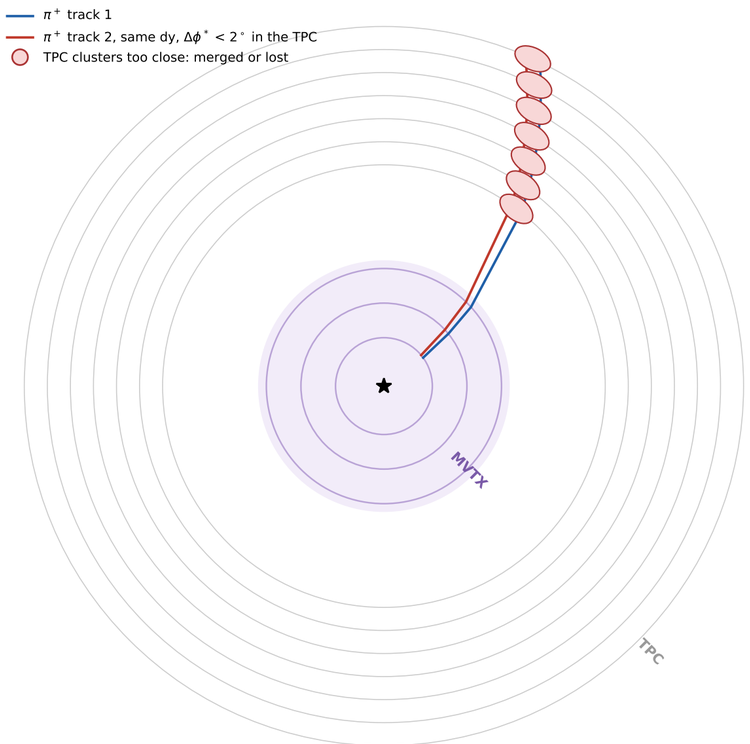


Without the veto: 3.8-4.05 at 20-35 MeV (ana578 3.6), on top of the rising Coulomb part (3.8, 2.1, 1.4 at 0-15 MeV). The veto removes the bump and leaves the Coulomb rise; the removed pairs reach to Q ~ 0.15 GeV (50-100 MeV: 1.27 -> 0.99). Like-sign within 0.01-0.03. 1+r_2 = total sibling / total mixed pairs (1.310 B1, 1.297 B4). "variable width binning".

Panel 10: the same-event case (ULS veto); the cross-crossing case is XTF (page 1, panel 2). Panel 11 is one particle (loser: fewer TPC clusters, then the larger |dca| sum). Numbers: README_SplitTracks581 secs B4, B5; panel 12: corral_ST581_B1/B4 ($|z_{vtx}|$ < 8 cm).

ana581, page 5: Areas C and D -- two real tracks, merged or lost (pair cuts), and the notch (watched) -- schematic, not to scale

13. C1 -- two like-sign tracks close all through the TPC
tsep: drop |dy| < 0.06 with min |dϕ^*| < 2.0° over R = 0.30-0.78 m

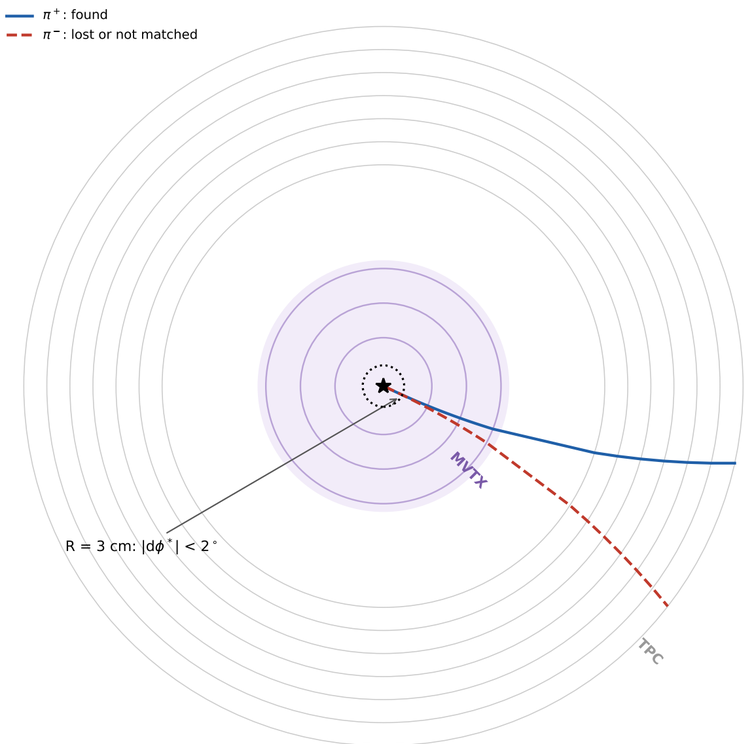


Two real tracks that stay within ~2° of each other through the TPC share or merge clusters: one is lost or both are damaged. Like-sign pairs at small Q do this (same curvature); unlike-sign pairs bend apart.

ana581 (all of 79516, tsep off): S/M 0.85 at min |dϕ^*| 0-0.5°, ~0.92 to 2° (|dy| < 0.06). LS C(Q)/(1+r_2): off is low below ~30 MeV; above ~15 MeV 2.0° = 3.0° (3.0° higher below 15 MeV, large errors). Decision: 2.0° stays (user).

Cut pairs from S and M alike: LS pions 0.36-0.37% S / 0.36-0.37% M (7 runs). Open: the dϕ = 0 dip in R_2 (|dy| < 0.02) left with either width (0.28 vs 0.36).

14. C2 -- two tracks that leave the vertex together
isep: drop |dy| < 1.0 with |dϕ^*(R = 3 cm)| < 2.0°, LS and ULS

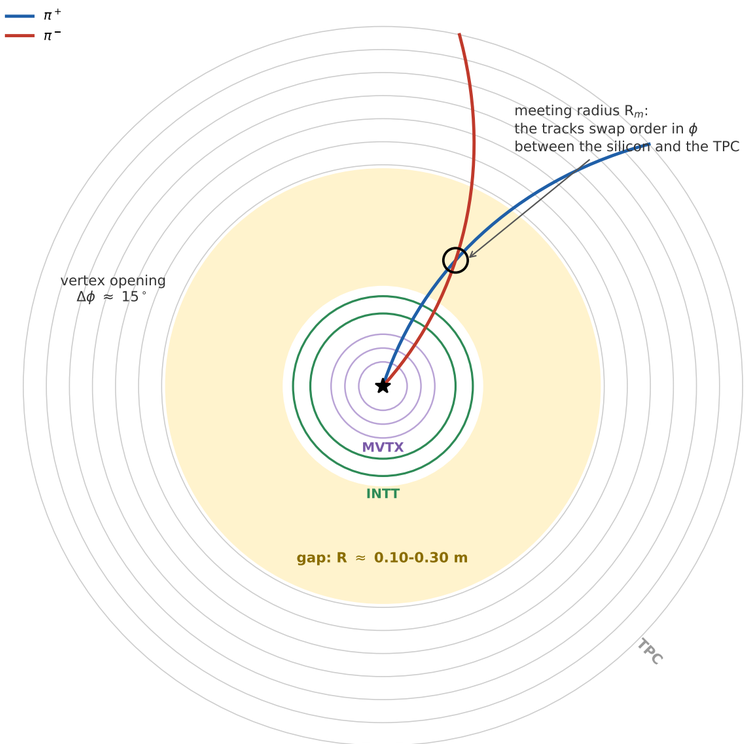


Two real tracks within ~2° at the innermost silicon share or confuse silicon clusters: one of the pair is often lost, like- or unlike-sign, at any dy up to ~1.

ana581 (all of 79516, isep off): S/M at |dϕ^*(3 cm)| 0-0.5°: LS 0.60 (|dy| < 0.4), ULS 0.75; back to 1 by 2° in every |dy| band, nothing left above 2°.

The missing same-event pairs inside a cut of width w peak at exactly w = 2.0°: a wider cut only removes loss-free pairs. 7 runs: ULS pions 0.80% S / 0.87% M, LS 0.82-0.83% / 0.86% (ana578 0.82/0.89, 0.84/0.88). Decision: 2.0° stays (user).

15. D1 -- the ±15° ULS notch: two tracks that meet in the INTT-TPC gap
LOST in the silicon-TPC matching -- watched, not removed



Opposite charges bend toward each other; a ~15° vertex opening puts the meeting point at $R_m \approx$ 0.10-0.30 m, where TPC seeds are matched to silicon.

ana581 (all of 79516): ULS depth ~0.98 (local pol2 baseline; not 1:1 with ana578); vs R_m 0.94 (0.10-0.15 m), 0.928, 0.93, 0.95 (0.25-0.30 m): as ana573 and ana578.

The loss is in pairs that MEET inside the gap (0.936; ana578 0.937) and does not shrink with dy (ULS 0.953 -> 0.913): the matching seems to use ϕ only. LS ~0.975, flat in R_m. Deepest at |dp_T| 0.2-0.3 GeV/c (0.92). No cut; monitored.

Panels 13-14: two real particles; the cut drops a region of pair space from sibling and mixed pairs alike (it cannot bring the lost pairs back). Panel 15: a loss upstream of Corral, monitored per production. Numbers: README_SplitTracks581 secs C1, C2, D1.